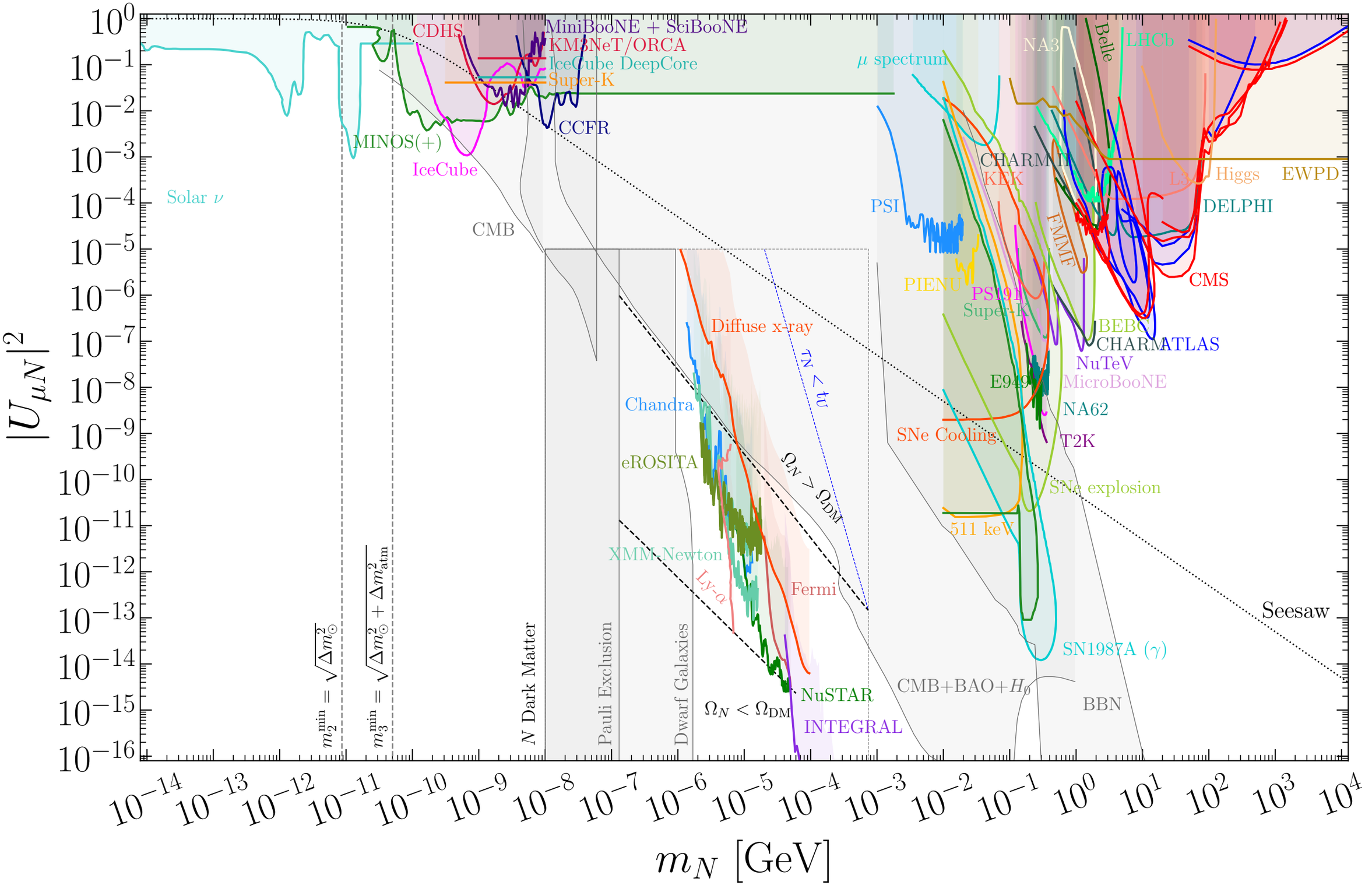

$|U_{\mu N}|^2$
m_N [GeV]
Solar ν
CDHS
MiniBooNE + SciBooNE
KM3NeT/ORCA
IceCube DeepCore
Super-K
μ spectrum
NA3
Belle
LHCb
MINOS(+)
CCFR
IceCube
CHARM II
KEK
L3
Higgs
EWPD
DELPHI
PSI
CMB
FMMF
PIENU
PS191
Super-K
CMS
BEBC
CHARM
ATLAS
NuTeV
Diffuse x-ray
$\tau_N < t_U$
E949
MicroBooNE
Chandra
NA62
T2K
SNe Cooling
eROSITA
$\Omega_N > \Omega_{\rm DM}$
SNe explosion
511 keV
XMM-Newton
Ly-α
Fermi
Seesaw
SN1987A (γ)
CMB+BAO+H_0
BBN
NuSTAR
$\Omega_N < \Omega_{\rm DM}$
INTEGRAL
$m_2^{\rm min} = \sqrt{\Delta m^2_\odot}$
$m_3^{\rm min} = \sqrt{\Delta m^2_\odot + \Delta m^2_{\rm atm}}$
N Dark Matter
Pauli Exclusion
Dwarf Galaxies